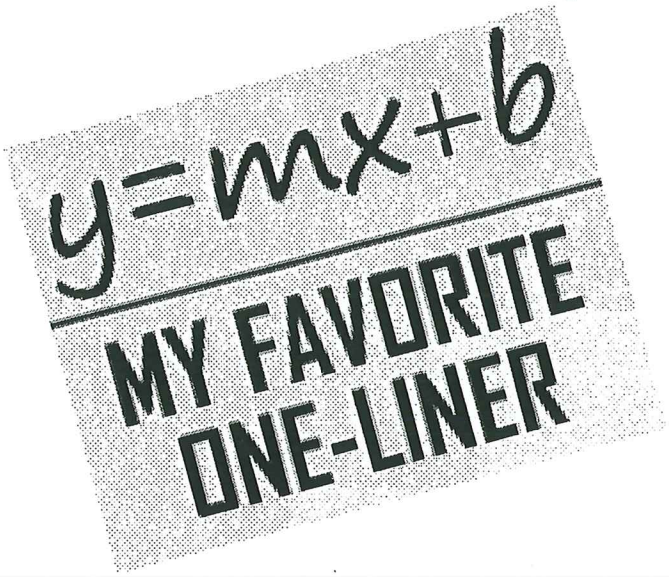
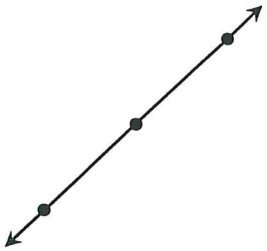
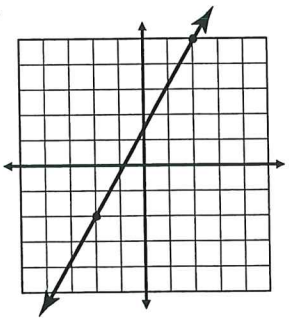
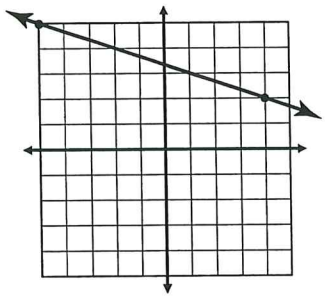
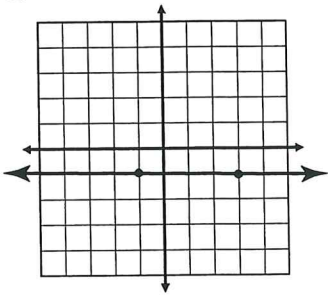
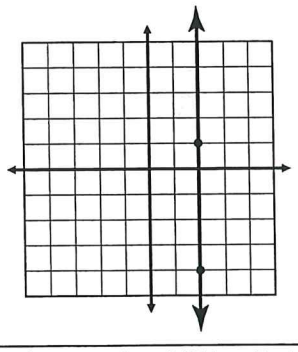
Algebra 1
Unit 2: Linear Equations



★ Rate of Change
★ Slope
★ Slope-Intercept Form
★ Standard Form
★ Graphing Linear Equations
★ x & y-intercepts
★ Vertical & Horizontal, Parallel & Perpendicular Lines
★ Writing Linear Equations
★ Point Slope Form
★ Word Problems

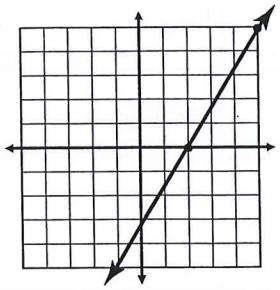
Name _____

Name:	Date:
Topic:	Class:

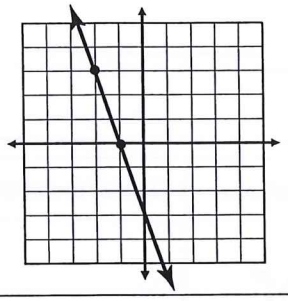
Main Ideas/Questions	Notes/Examples
Rate of Change	On a linear graph, this is called the _____ of the line!
Slope 	• Slope is written as a _____ of the **vertical change** (_____) to the **horizontal change** (_____) between any two points on a line. • This remains _____ for any two points on the same line. • Slope is written as a _____ in _____ _____. • Variable for slope: _____
Types of Slope	
Finding Slope on a Graph m =	**Directions:** Find the slope of each line. Write your answer in simplest form! 1.  2.  3.  4. 

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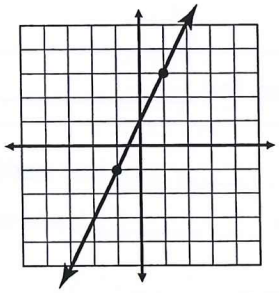
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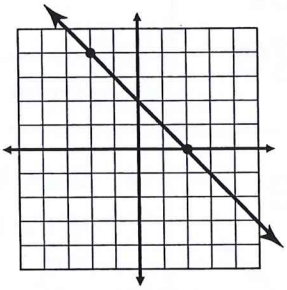
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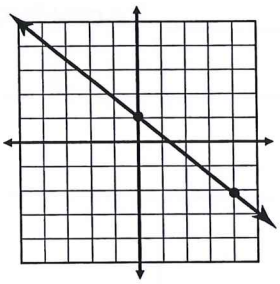
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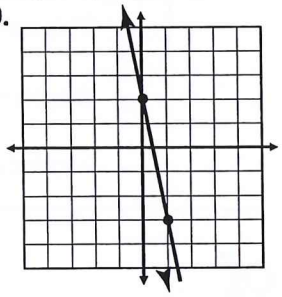
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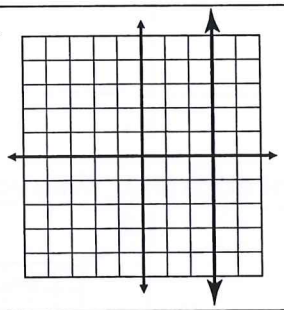
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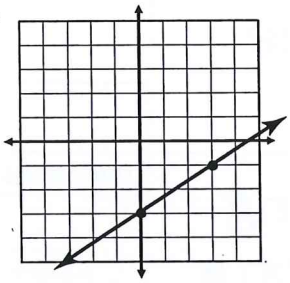
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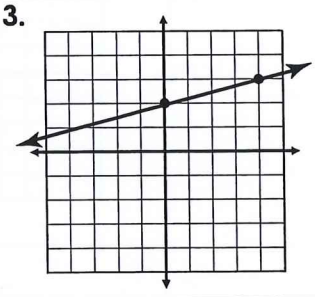
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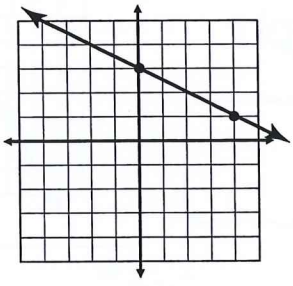
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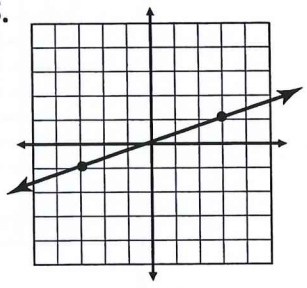
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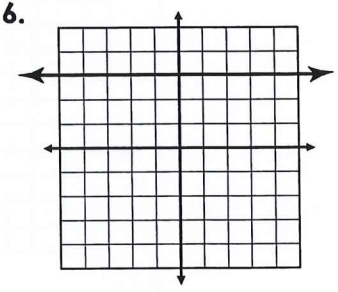
14.



15.



16.



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2

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
SLOPE FORMULA	The **slope formula** is used to find the slope between two points (x_1, y_1) and (x_2, y_2).	
	Formula:	
	It is important to remember to SIMPLIFY your answer!	
EXAMPLES	**Directions:** Find the slope between each pair of points.	
	1. (1, 1) and (4, 3)	**2.** (-2, 4) and (10, -2)
	3. (-4, 5) and (-8, -5)	**4.** (10, 0) and (-2, 4)
	5. (5, 9) and (3, 9)	**6.** (-7, 8) and (-7, 5)
	7. (-1, 9) and (2, 3)	**8.** (-4, 13) and (6, -2)
	9. (5, 6) and (6, 5)	**10.** (9, -4) and (1, -4)

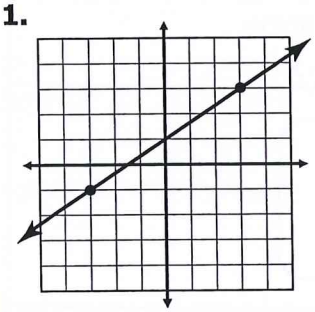
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Name: _____ **Unit 4:** Linear Equations

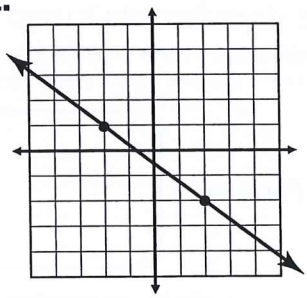
Date: _____ **Bell:** _____ **Homework 1:** Slope

Directions: Find the slope of the line show on the graph.

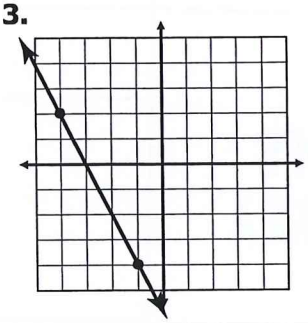
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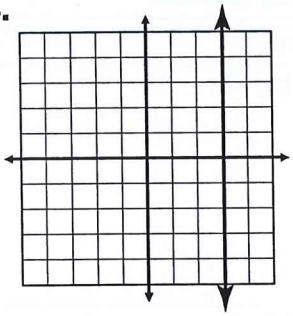
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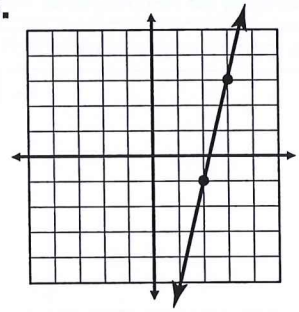
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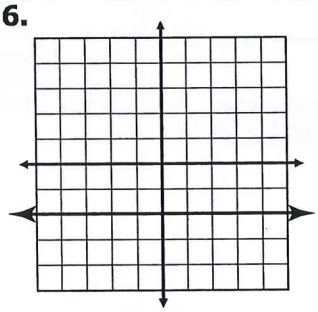
4.



5.



6.



Directions: Use the slope formula to find the slope between the given two points.

7. (4, 9) and (1, 6)

8. (-4, -1) and (-2, -5)

9. (5, 3) and (5, -9)

10. (2, 1) and (8, 9)

11. (14, -8) and (7, -6)

12. (4, -3) and (8, -3)

Directions: Find the missing value so that the line passing through the points has the given slope.

13. $(x, -4)$ and (2, 8); $m = -3$

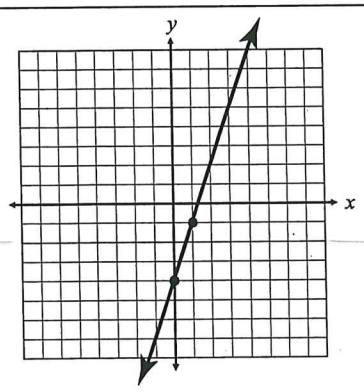
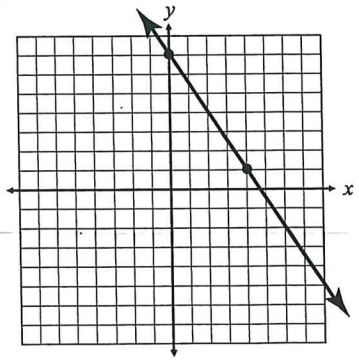
14. $(10, y)$ and (3, 4); $m = -\frac{2}{7}$

15. (7, 1) and $(x, 4)$; $m = \frac{3}{4}$

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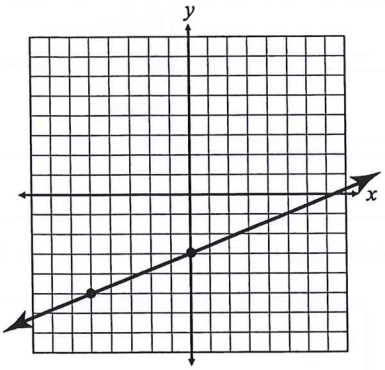
4

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
Slope-Intercept Form	Linear equations are frequently written in **slope-intercept form**: $y = mx + b$ m is the slope and b is the y-intercept
Examples	**Directions:** Given the slope and y-intercept of the line, write the equation in slope-intercept form. **1.** slope = 2; y-intercept = -1 _____ **2.** slope = $-\frac{3}{5}$; y-intercept = 4 _____ **3.** slope = -3; y-intercept = 2 _____ **4.** slope = -1; y-intercept = 7 _____ **5.** slope = $\frac{1}{4}$; y-intercept = 0 _____ **6.** slope = $-\frac{5}{2}$; y-intercept = -3 _____
Given a Graph	**Directions:** Identify the slope and y-intercept of the line on the graph. Then, write the equation of the line in slope-intercept form. **7.**  m = _____ b = _____ Equation: _____ **8.**  m = _____ b = _____ Equation: _____

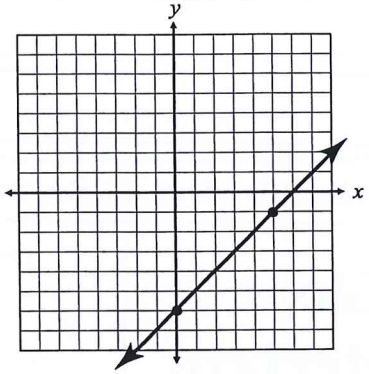
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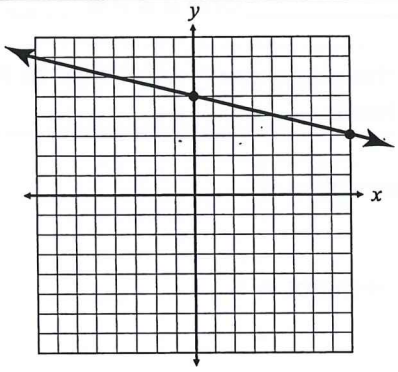
Equation: _____

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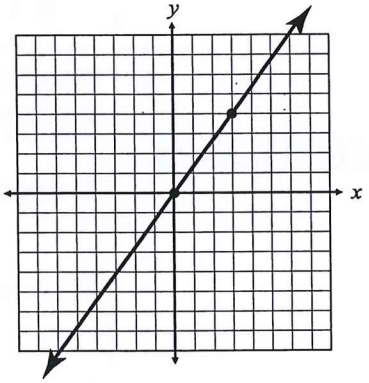
Equation: _____

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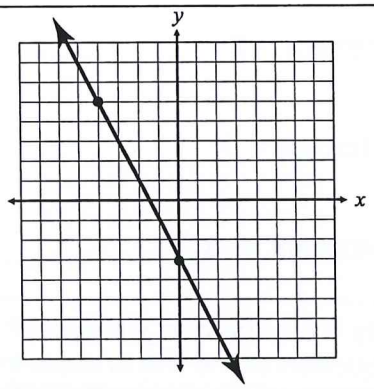
Equation: _____

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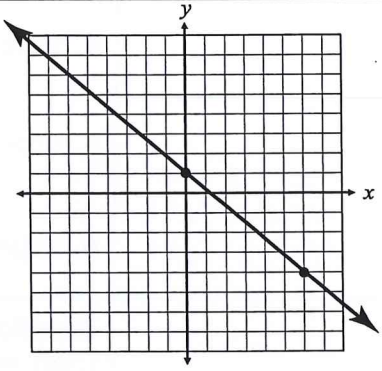
Equation: _____

13.



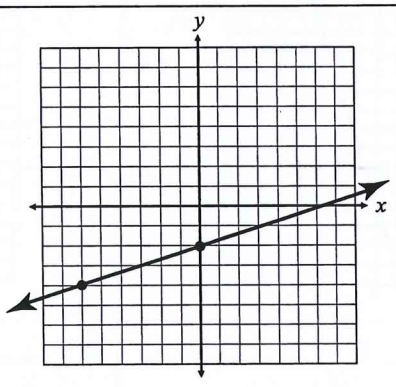
Equation: _____

14.



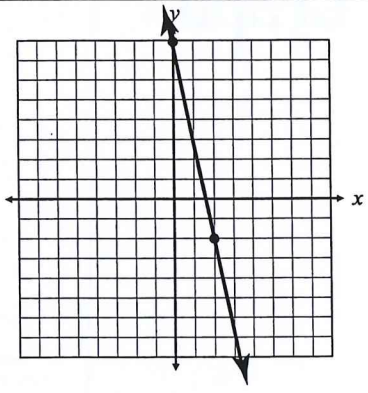
Equation: _____

15.



Equation: _____

16.



Equation: _____

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
STANDARD FORM	Linear equations are also frequently written in **standard form**: 
CONVERTING Standard Form ➔ Slope-Intercept Form	**Equations written in standard form can be converted to slope-intercept form by solving for y!**

Directions: Write each equation in slope-intercept form. Identify the slope and y-intercept.

1. $2x + y = 3$	**2.** $4x + 5y = -30$
3. $x - 3y = 12$	**4.** $x - y = -8$
5. $4x - y = 0$	**6.** $3x - 2y = -14$
7. $-6x + 4y = 20$	**8.** $x + 5y = -15$

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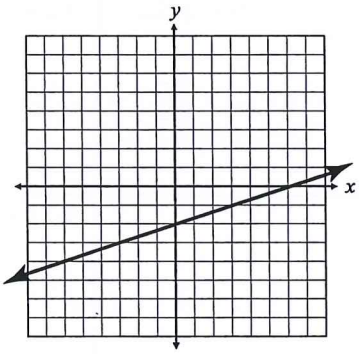
7

	9. $3x + 3y = -12$	**10.** $-2x - 4y = 36$
	11. $12x - 9y = -9$	**12.** $4x + 10y = 10$

GIVEN GRAPHS

Directions: Determine the equation that best represents the line shown on the graph. ***Hint: Convert them to slope-intercept form!**

13.



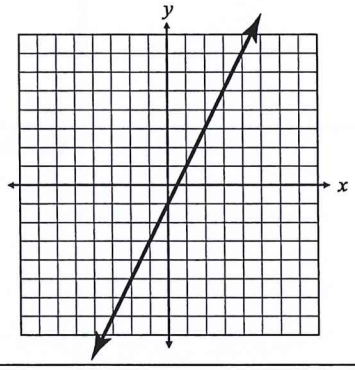
A. $x + 3y = -6$

B. $x - 3y = 6$

C. $3x + y = -2$

D. $3x - y = 2$

14.



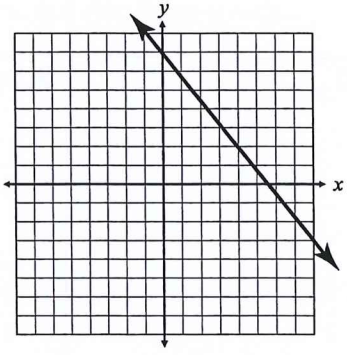
A. $x + 2y = -2$

B. $x - 2y = 2$

C. $2x + y = -1$

D. $2x - y = 1$

15.



A. $4x + 5y = 35$

B. $4x - 5y = -35$

C. $5x + 4y = 28$

D. $5x - 4y = -28$

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Name: _____ **Unit 4:** Linear Equations

Date: _____ **Bell:** _____ **Homework 2:** Slope-Intercept & Standard Form

**** This is a 2-page document! ****

Directions: Give the equation for each form below.

Slope-Intercept Form	**Standard Form**

Directions: Write the equation of the line with the given slope and y-intercept in slope-intercept form.

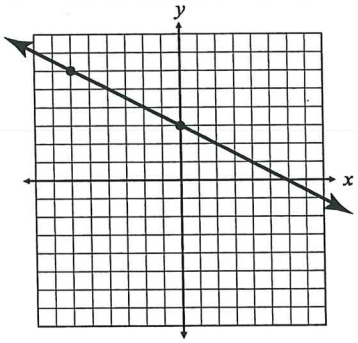
1. slope = $\frac{1}{3}$; y-intercept = -1

2. slope = $-\frac{2}{5}$; y-intercept = 0

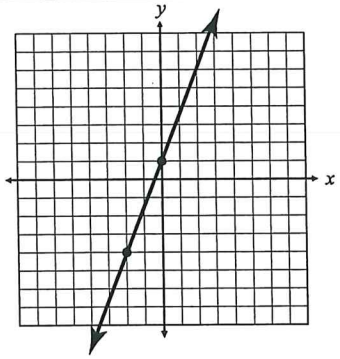
3. slope = 1; y-intercept = 8

Directions: Write the equation of the line shown on the graph in slope-intercept form.

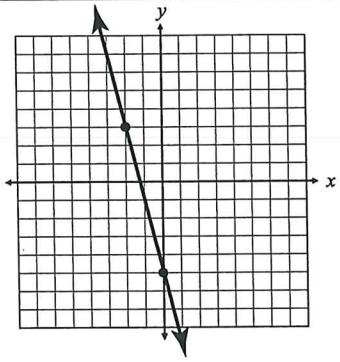
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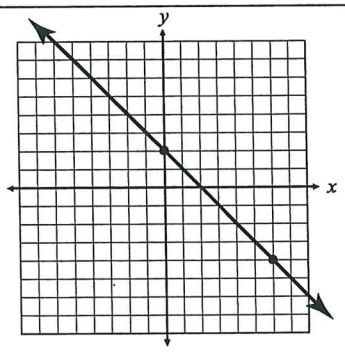
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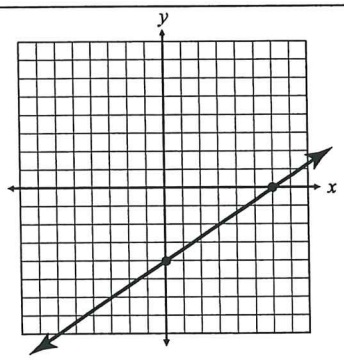
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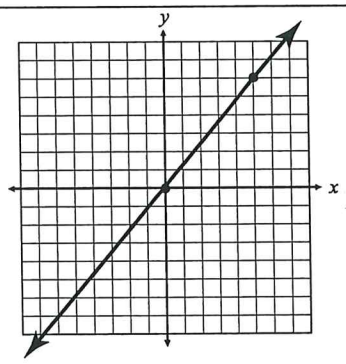
7.



8.



9.



Directions: Convert the following equations from standard form to slope-intercept form.

10. $6x + 2y = 10$	**11.** $x + 2y = 4$

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12. $x + y = -3$

13. $-2x + 3y = -6$

14. $4x + y = -1$

15. $5x - 2y = 14$

16. $10x - 2y = -6$

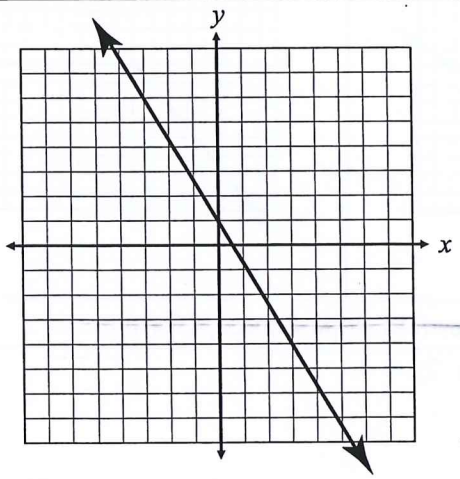
17. $8x + 6y = 12$

18. $6x - 10y = 20$

19. $3x + 9y = -36$

Directions: Determine the equation that best represents the line shown on the graph.

20.



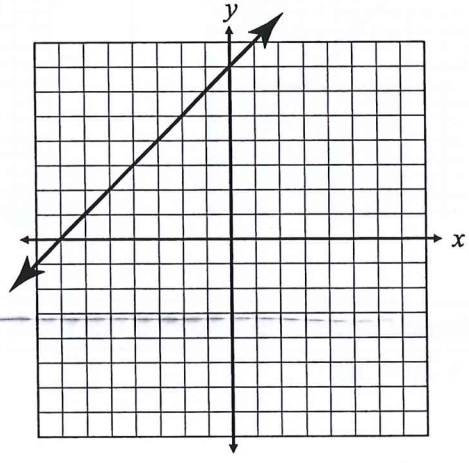
A. $-5x + 3y = 3$

C. $5x + 3y = 3$

B. $5x + 3y = -3$

D. $5x - 3y = -3$

21.



A. $x - y = -7$

C. $x - y = 7$

B. $x + y = -7$

D. $x + y = 7$

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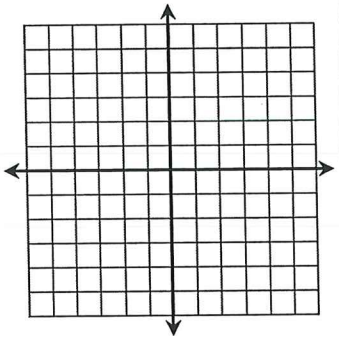
10

Name:	Date:
Topic:	Class:

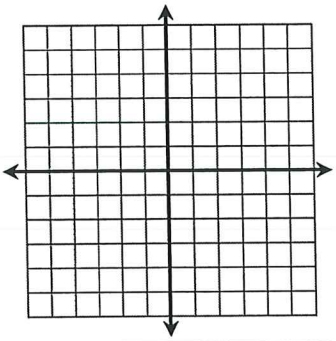
Main Ideas/Questions	Notes/Examples
GRAPHING LINEAR EQUATIONS (By Slope-Intercept)	**Use the steps below to graph an equation using slope-intercept form:**
	❶ Write the equation in **slope-intercept form.**
	❷ Graph the ***y*-intercept.** This is always point $(0, b)$.
	❸ Use the **slope** of the line to create more points. Remember slope is rise/run!
	❹ Use a ruler to draw a line that extends through the points, placing an arrow on both ends.

Directions: Graph each equation using the slope-intercept method.

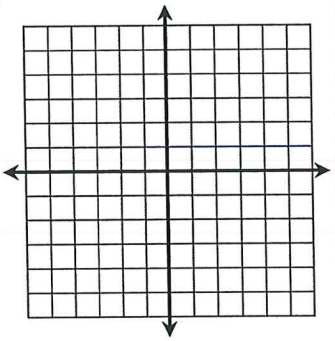
1. $y = -x + 5$



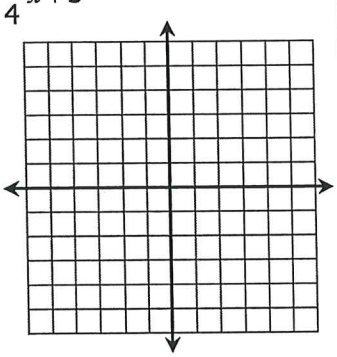
2. $y = -3x - 1$



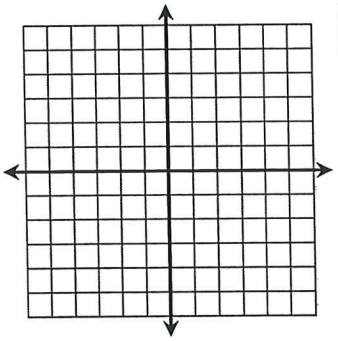
3. $y = \frac{2}{5}x + 2$



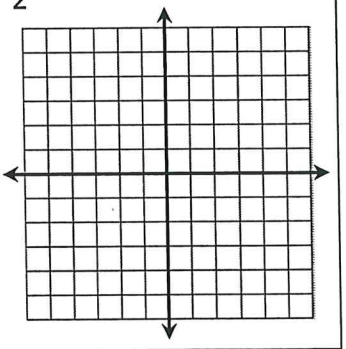
4. $y = -\frac{1}{4}x + 3$



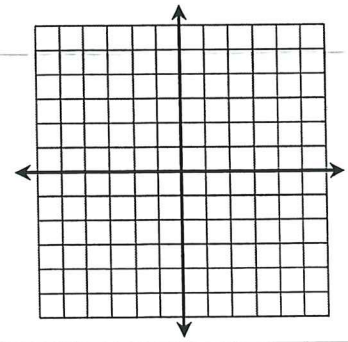
5. $y = 2x + 6$



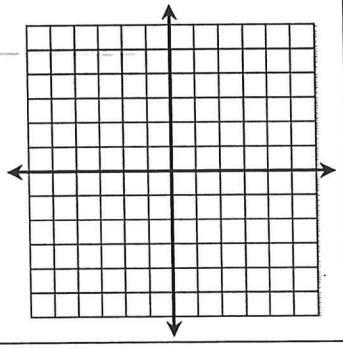
6. $y = -\frac{3}{2}x - 5$



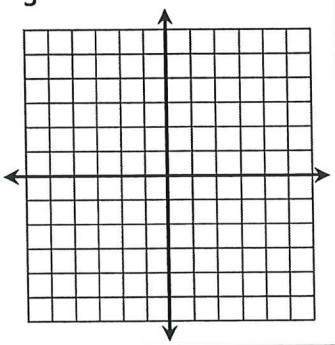
8. $y = -4x$



7. $y = -3 + 5x$

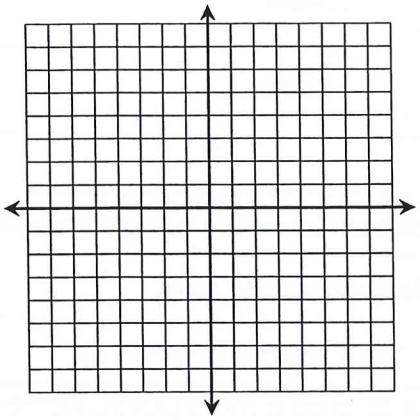


9. $y = 1 - \frac{6}{5}x$

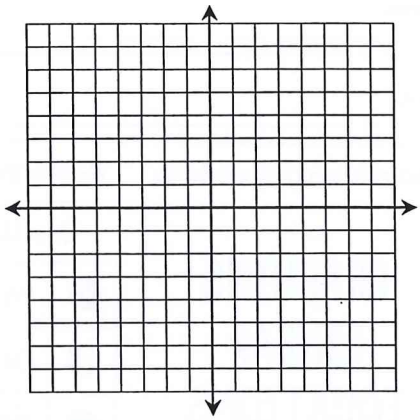


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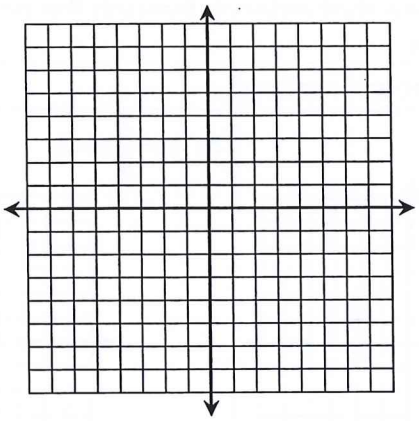
10. $x + 3y = -3$



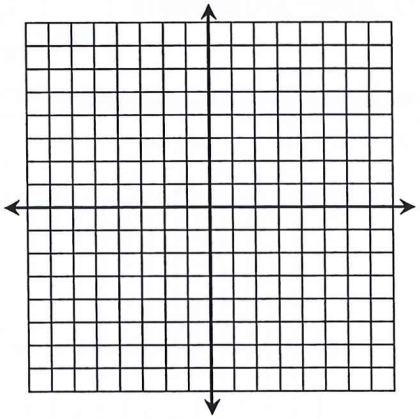
11. $x - y = -5$



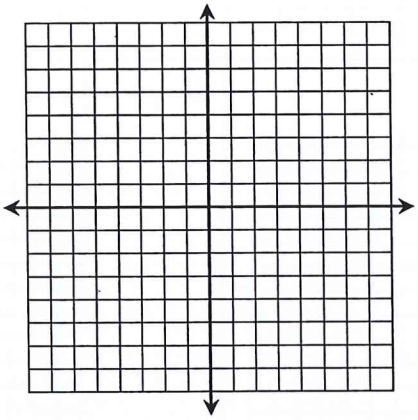
12. $5x - y = -3$



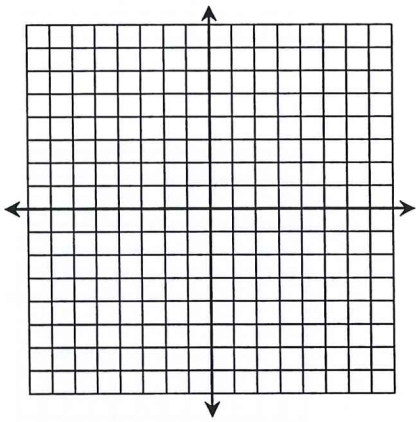
13. $4x + 3y = 21$



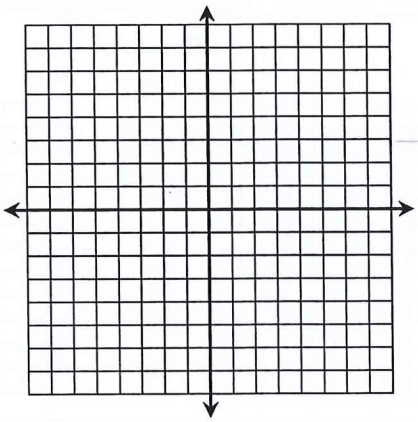
14. $2x - 4y = 20$



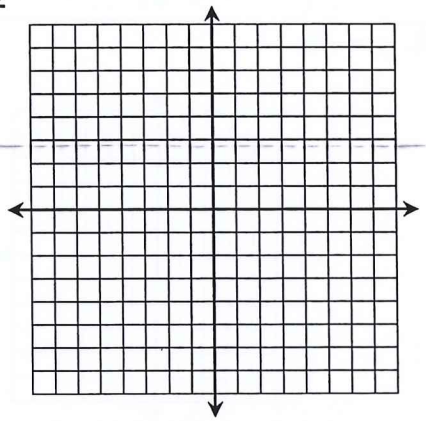
15. $2x + 3y = 0$



16. $10x - 8y = 24$



17. $9x + 12y = 12$



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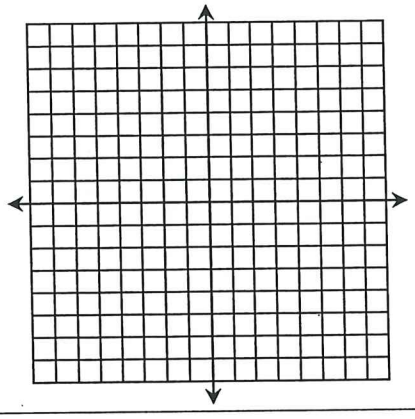
Name: _____ **Unit 4:** Linear Equations

Date: _____ **Bell:** _____ **Homework 3:** Graphing Linear Equations by Slope-Intercept Form

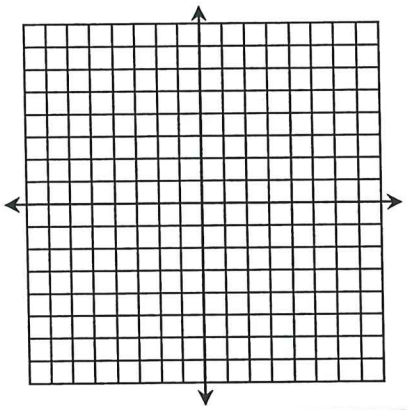
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Directions: Graph the following linear equations. Convert to slope-intercept form when necessary.

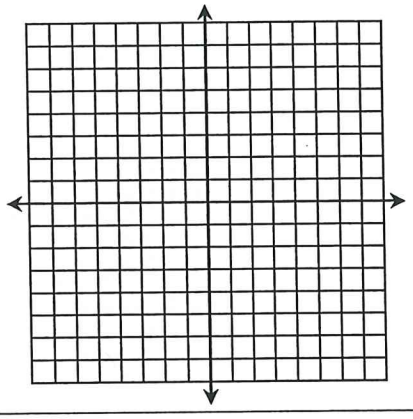
1. $y = -\frac{2}{5}x - 3$



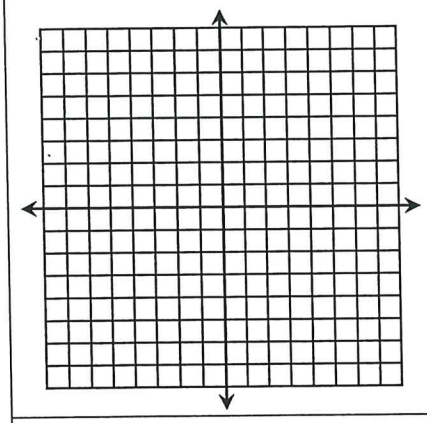
2. $y = \frac{1}{2}x + 5$



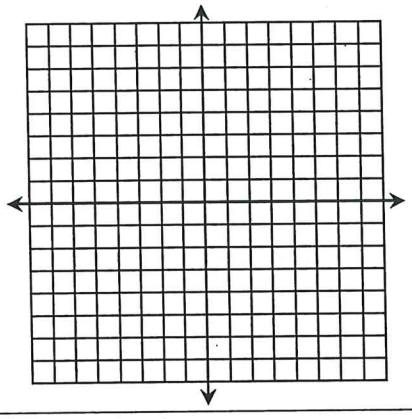
3. $y = 6x - 1$



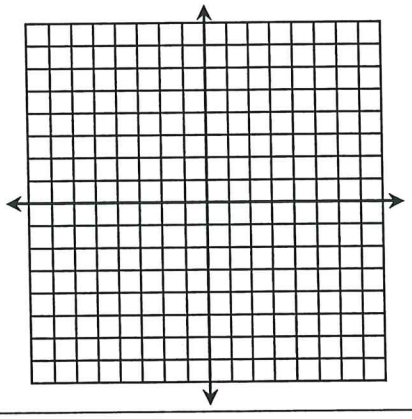
4. $y = -x + 6$



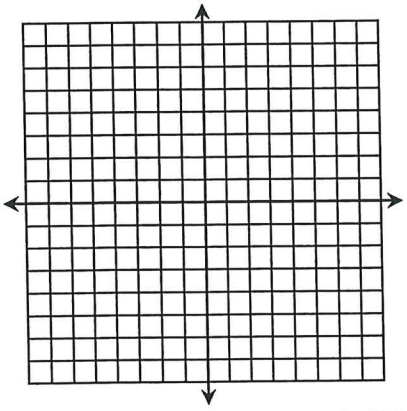
5. $y = -\frac{3}{4}x$



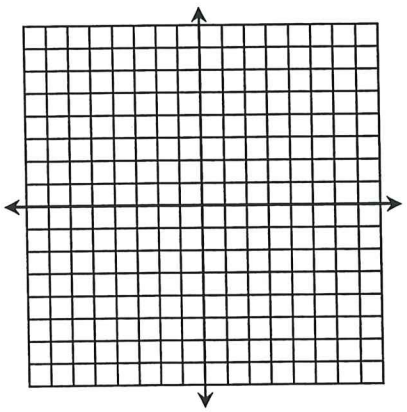
6. $y = \frac{5}{4}x - 3$



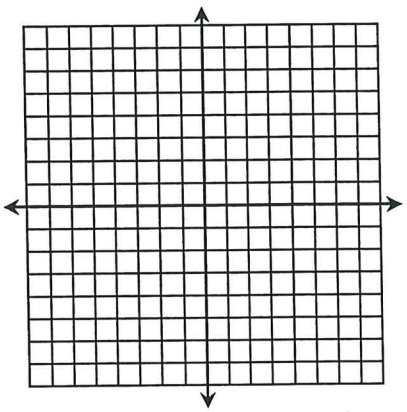
7. $y = 7 - 5x$



8. $y = -4 + \frac{6}{5}x$

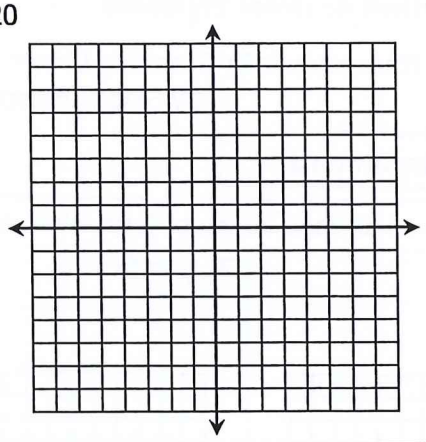


9. $y = -5 - \frac{1}{3}x$

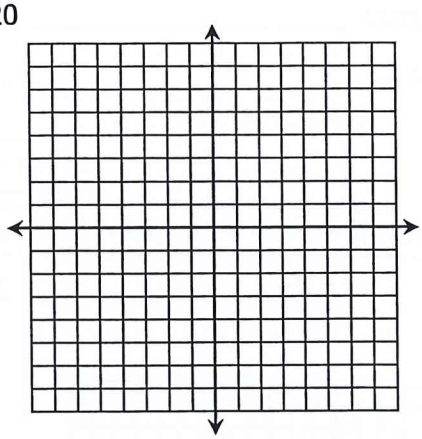


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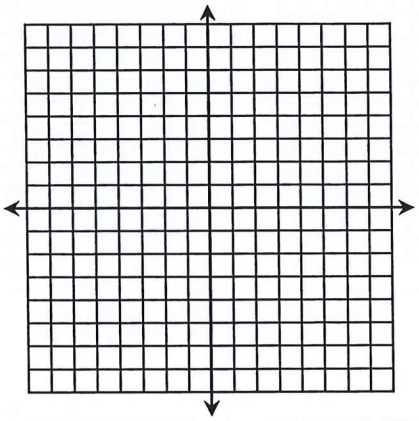
10. $5x + 4y = -20$



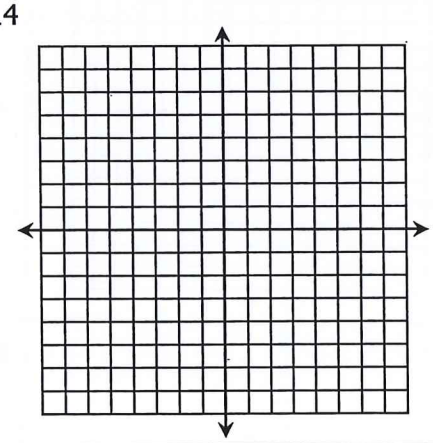
11. $2x - 10y = 20$



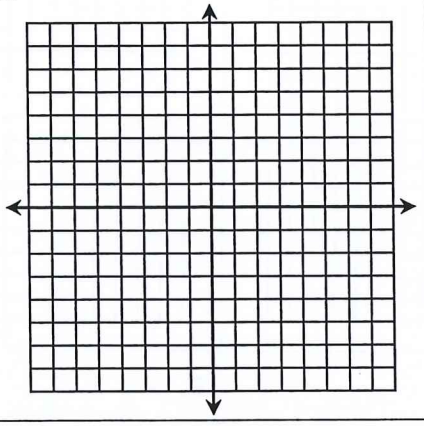
12. $x - y = 0$



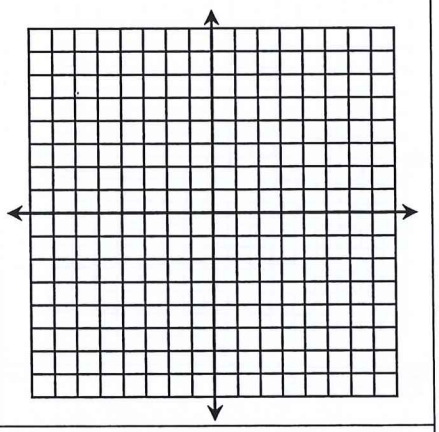
13. $6x + 2y = 14$



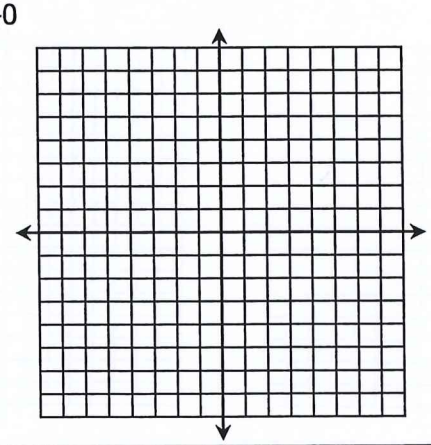
14. $3x + 4y = -12$



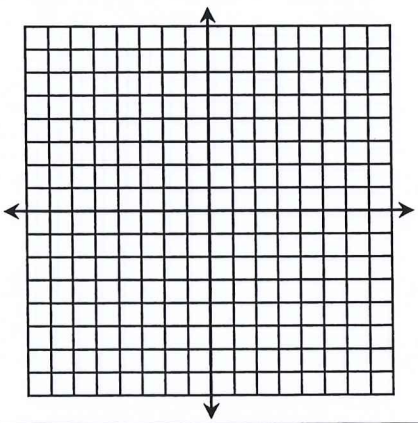
15. $4x - y = 1$



16. $12x - 8y = 40$

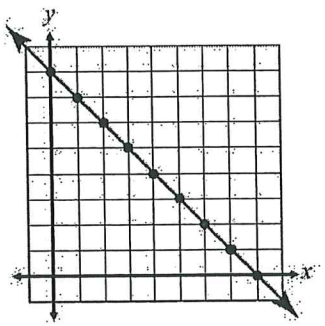


17. $7x + 2y = 8$



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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
X- and Y-Intercepts	➢ The point at which the line intersects the x-axis is called the ***x*-intercept.** ➢ The point at which the line intersects the y-axis is called the ***y*-intercept.** ➢ **Example:** Identify the x- and y-intercept of the graph shown to the right. 
Finding Intercepts Algebraically	➢ To find the x-intercept of an equation: _____ ➢ To find the y-intercept of an equation: _____ ➢ **Example:** Find the x- and y-intercept of the equation $y = 3x + 6$.
Examples	**Directions:** Find the x- and y-intercept of each equation. **1.** $y = -x + 5$ ***x*-int:** _____ ***y*-int:** _____ **2.** $y = \frac{1}{2}x - 8$ ***x*-int:** _____ ***y*-int:** _____ **3.** $y = -\frac{4}{3}x + 2$ ***x*-int:** _____ ***y*-int:** _____

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4. $x - y = 2$

x-int: _____

y-int: _____

5. $3x - 2y = 12$

x-int: _____

y-int: _____

6. $8x + 10y = -10$

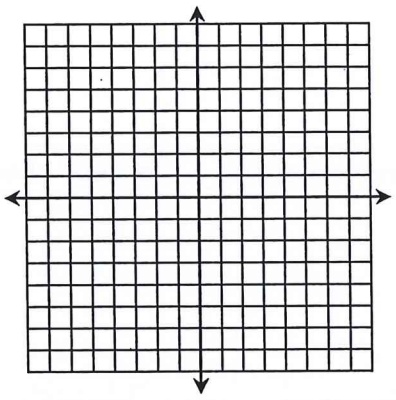
x-int: _____

y-int: _____

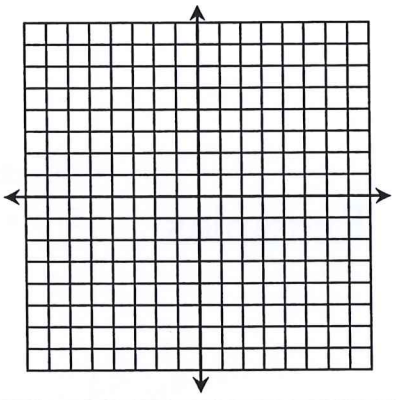
Graphing by Intercepts

Directions: Find the x- and y-intercept of each equation. Graph the equation using its intercepts.

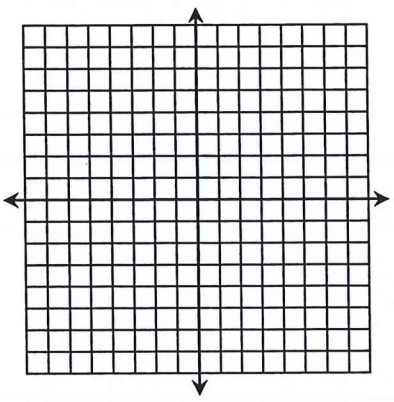
7. $x + y = 3$



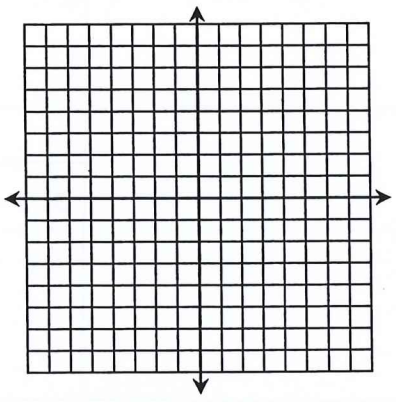
8. $-4x + 5y = 20$



9. $9x - 15y = 45$



10. $2x - y = 7$



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Name: _____ **Unit 4:** Linear Equations

Date: _____ **Bell:** _____ **Homework 4:** x- and y-Intercepts

**** This is a 2-page document! ****

Directions: Find the x-intercept and y-intercept of each equation algebraically.

1. $y = -2x + 6$

x-int: _____

y-int: _____

2. $y = x - 7$

x-int: _____

y-int: _____

3. $y = \frac{3}{2}x - 9$

x-int: _____

y-int: _____

4. $y = -\frac{1}{4}x + 1$

x-int: _____

y-int: _____

5. $-x + y = -5$

x-int: _____

y-int: _____

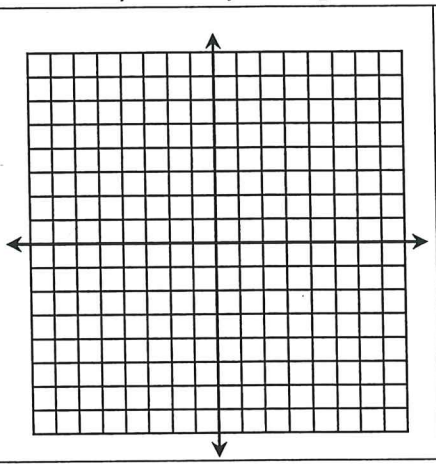
6. $x + 2y = -14$

x-int: _____

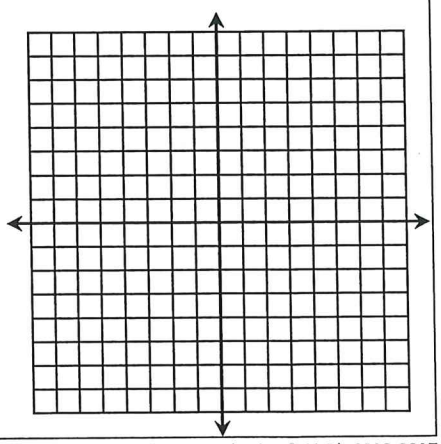
y-int: _____

Directions: Graph each equation by finding its x-intercept and y-intercept.

7. $3x - y = -3$



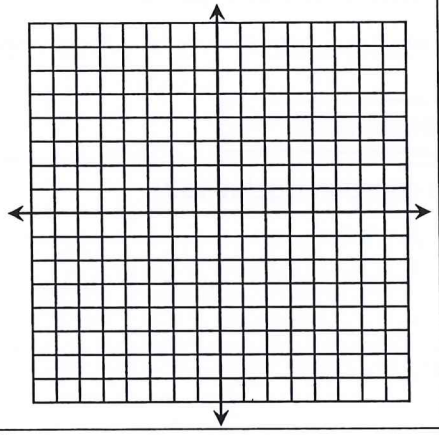
8. $5x + 3y = 15$



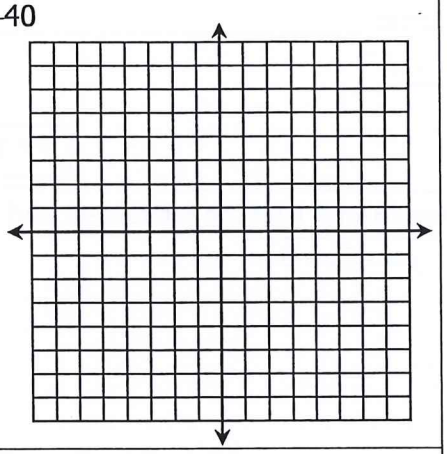
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17

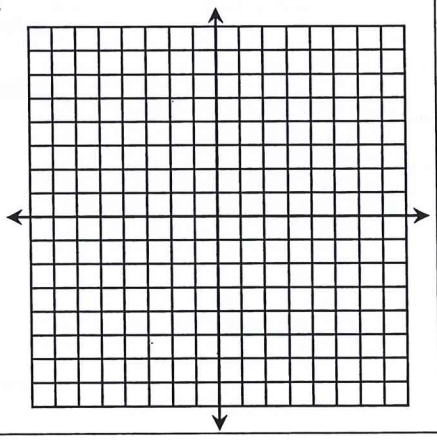
9. $x - 4y = 8$



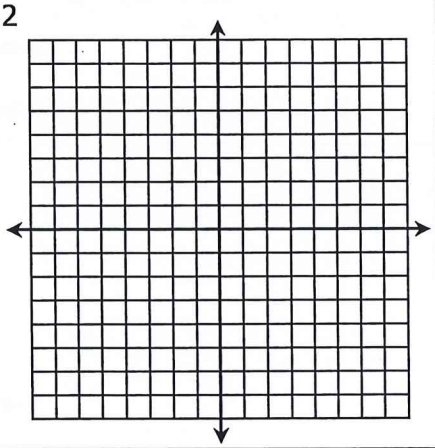
10. $10x + 8y = -40$



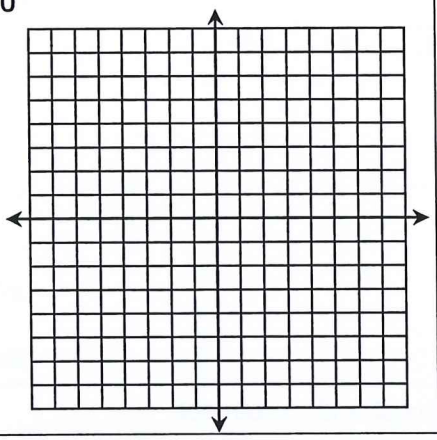
11. $4x - 6y = 12$



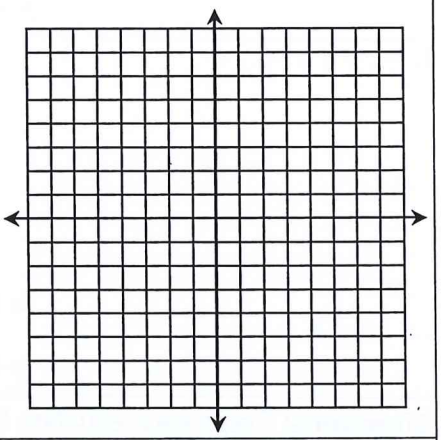
12. $-3x + 6y = 12$



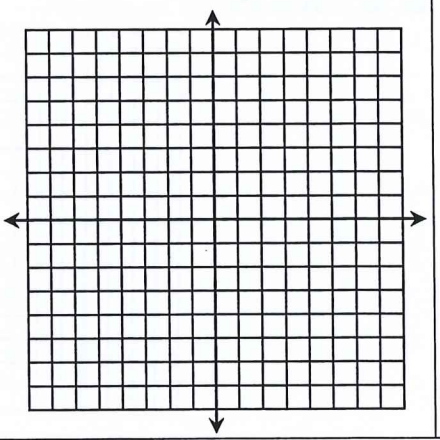
13. $-2x - 5y = 10$



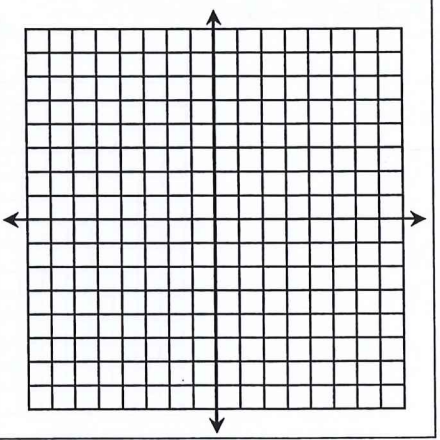
14. $x - y = 1$



15. $2x + 3y = 9$

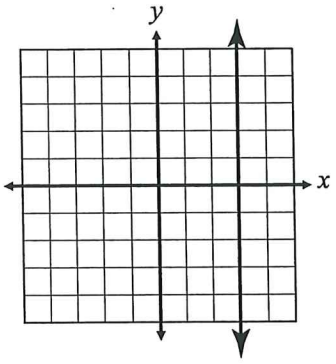
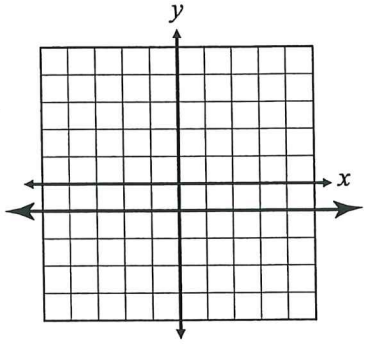


16. $5x - y = -2$

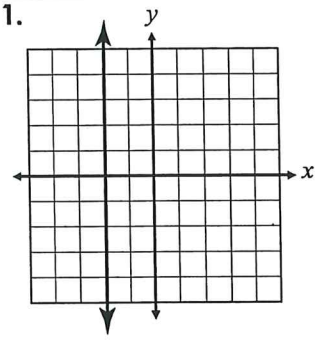


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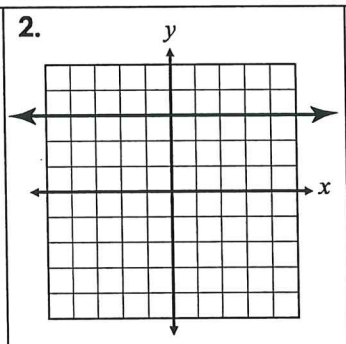
Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
Vertical & Horizontal Lines	**Vertical Lines**	**Horizontal Lines**
	A **vertical line** is written in the form $x = a$, where a represents the line's x-intercept.  The equation of the vertical line graphed above is _____	A **horizontal line** is written in the form $y = a$, where a represents the line's y-intercept.  The equation of the **horizontal** line graphed above is _____
	Remember, if the line intersects the **x-axis**, it's $x = a$, if a line intersects the **y**-axis**, it's $y = a$.**	
Examples	**Directions:** Write the equation of the line shown on the graph.	

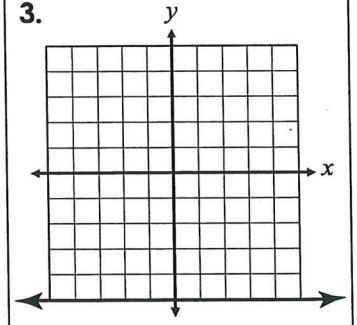
1.



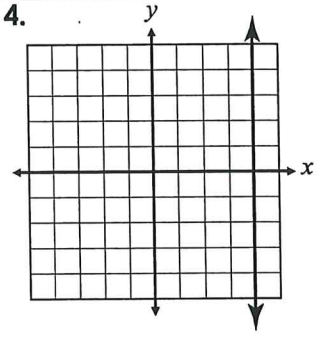
2.



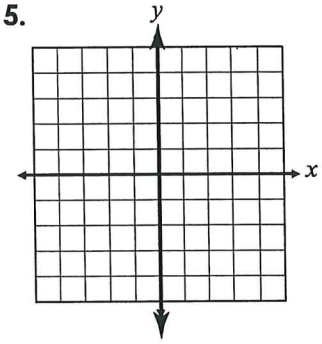
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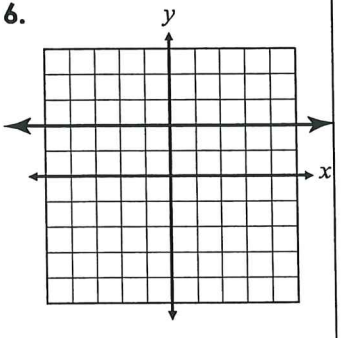
4.



5.



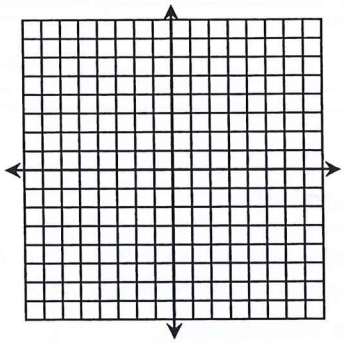
6.



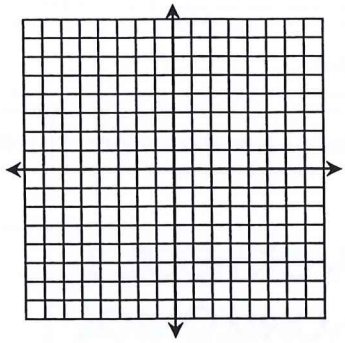
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Directions: Graph each equation.

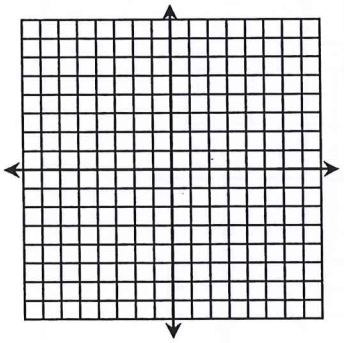
7. $y = 5$



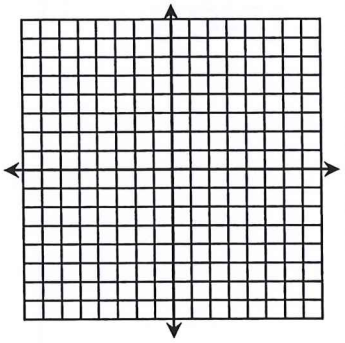
8. $x = -1$



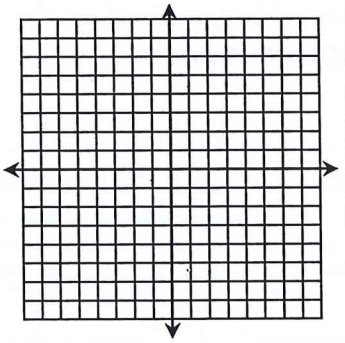
9. $x = 3$



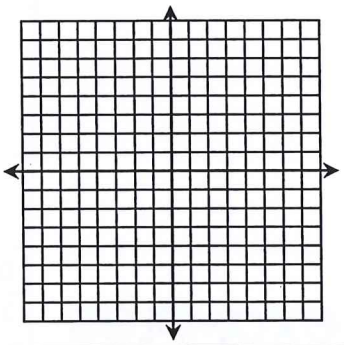
10. $y = -2$



11. $y = 4$



12. $x = -7$



QUESTIONS

13. What is the slope of the line $y = 4$?

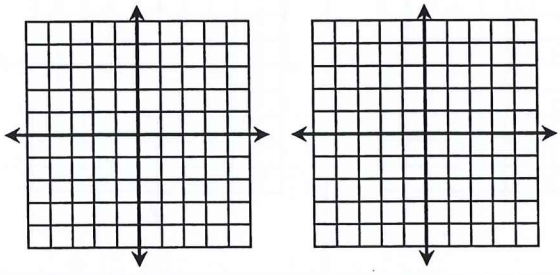
14. What is the slope of the line $x = -2$?

15. What is the slope of the line $x = 0$?

16. Which axis is $y = -1$ parallel to?

17. Which axis is $x = 4$ parallel to?

18. How does the graph of $y = 3$ differ from $y = 3x$? Graph both and explain.



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Name: _____ **Unit 4:** Linear Equations

Date: _____ **Bell:** _____ **Homework 5:** Slope & Graphing Lines Review

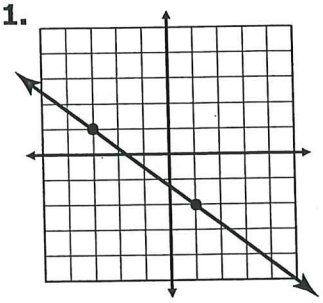
**** This is a 2-page document! ****

Directions: Identify each of the following below.

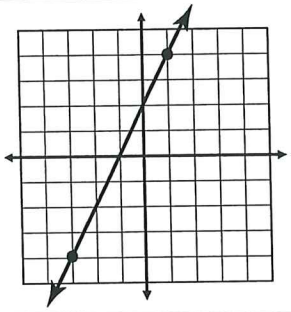
SLOPE FORMULA	SLOPE-INTERCEPT FORM	STANDARD FORM

Directions: Find the slope of the line shown on the graph.

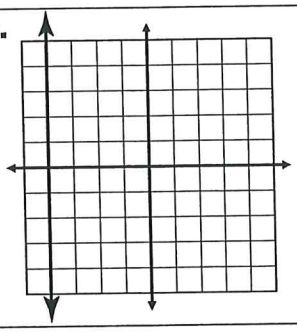
1.



2.



3.



Directions: Find the slope of the line that passes through the given two points.

4. (1, -4) and (-4, -6)

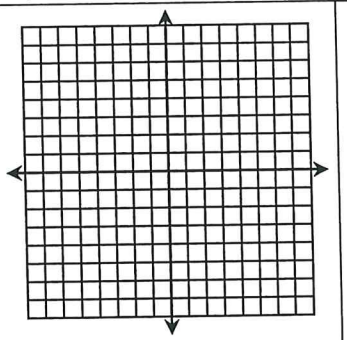
5. (-6, 2) and (-5, -2)

6. (-3, 3) and (-9, 5)

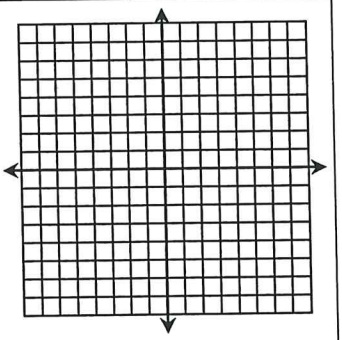
7. (4, -7) and (2, -7)

Directions: Graph each equation using slope-intercept form.

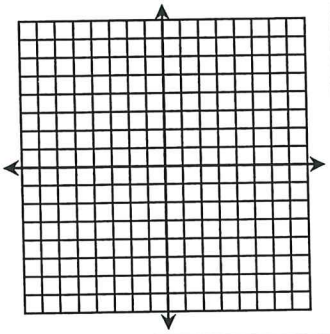
8. $y = x + 6$



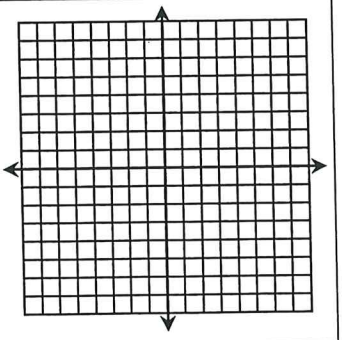
9. $y = -\frac{5}{3}x + 1$



10. $y = \frac{1}{2}x - 5$



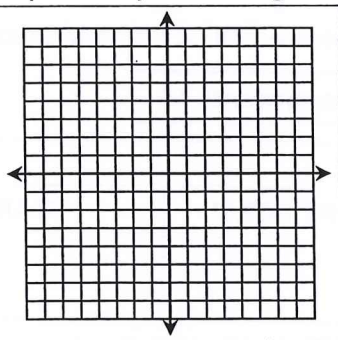
11. $y = -2 - 4x$



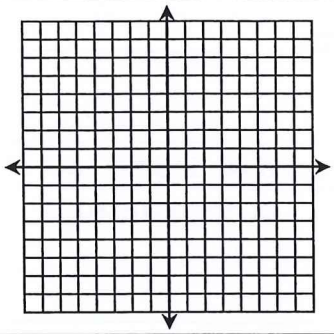
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Directions: Graph each equation by converting to slope-intercept form. **SHOW ALL WORK!**

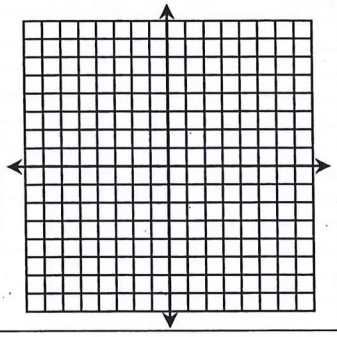
12. $6x - 2y = -14$



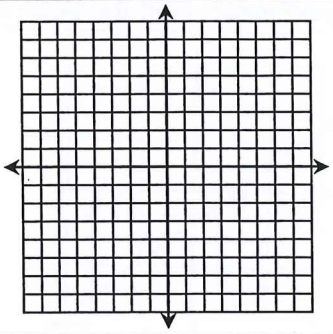
13. $x - y = -3$



14. $x - 4y = -24$

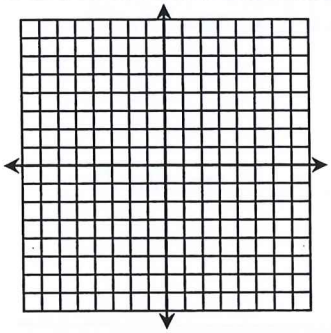


15. $10x + 4y = 0$

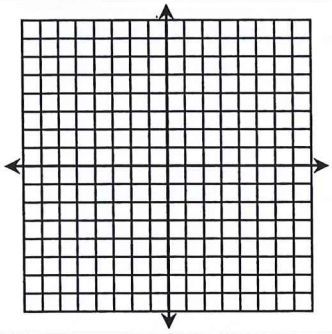


Directions: Graph each equation by finding its x- and y-intercept. **SHOW ALL WORK!**

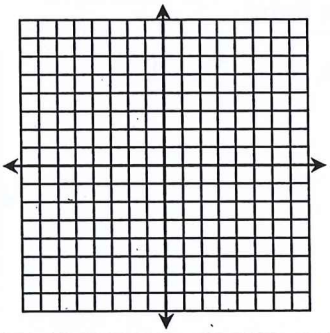
16. $x - y = 5$



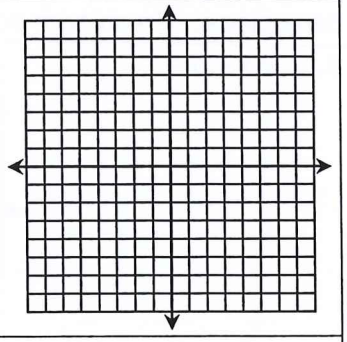
17. $x + 5y = -5$



18. $9x + 12y = -36$

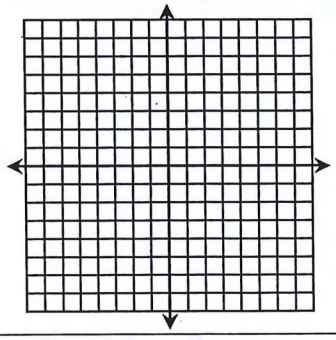


19. $16x - 10y = 20$

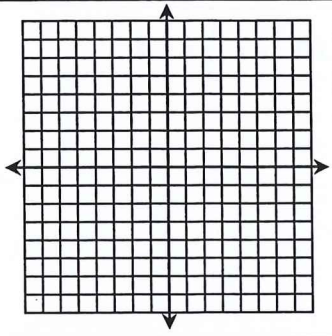


Directions: Graph each equation.

20. $y = 1$



21. $x = -6$



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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
WRITING LINEAR EQUATIONS (Given a Point and Slope)	To write the equation of the line passing through point (x_1, y_1) with slope (m), you can use the point-slope formula: **Point-Slope Formula:** *Be sure to distribute and solve for y!*
EXAMPLES	**Directions:** Write the equation of the line that passes through the given point and has the given slope. Write all final answers in slope-intercept form.

1. (4, 1); slope = 2	**2.** (2, 4); slope = $\frac{1}{2}$
3. (-6, 0); slope = $\frac{2}{3}$	**4.** (-8, -1); slope = $-\frac{3}{4}$
5. (4, -3); slope = -1	**6.** (0, -9); slope = 4

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	7. (-12, -5); slope = $\frac{1}{3}$	**8.** (3, -8); slope = -3
	9. (5, -9); slope = $-\frac{2}{5}$	**10.** (-4, -7); slope = 1
	11. (-6, -1); slope = $-\frac{4}{3}$	**12.** (4, 9); slope = 5
	13. (3, -4); slope = 0	**14.** (-6, -8); slope = $\frac{5}{6}$
	15. (-4, 1); slope = $-\frac{1}{4}$	**16.** (7, -8); slope = -2

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
WRITING LINEAR EQUATIONS (Given Two Points)	To write a linear equation that passes through two points, (x_1, y_1) and (x_2, y_2), use the slope formula followed by the point-slope formula: **Slope Formula** → **Point-Slope Formula**
EXAMPLES	**Directions:** Write the equation of the line that passes through the given two points. Write all final answers in slope-intercept form.

1. (-3, 7) and (1, -1)

2. (-6, -7) and (3, -4)

3. (2, -1) and (4, -6)

4. (1, 6) and (2, 5)

5. (-8, 1) and (0, -5)

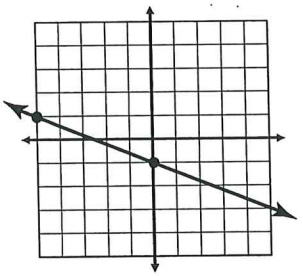
6. (-3, -8) and (2, 7)

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	7. (-9, -14) and (3, 2)	**8.** (-6, -3) and (-4, -1)
	9. (-4, 7) and (6, 2)	**10.** (1, 1) and (-1, -9)
	11. (4, -1) and (0, 2)	**12.** (-2, -2) and (1, -5)
	13. (3, -5) and (2, -1)	**14.** (-3, 14) and (2, -1)

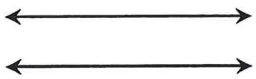
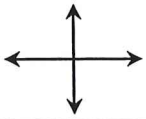
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I can write Linear Equations in SLOPE-INTERCEPT FORM GIVEN:

The Slope and y-Intercept slope = $\frac{1}{3}$; y-intercept = -5	
A Graph 	
An Equation in Standard Form $4x - 2y = 14$	
A Point and Slope (-1, 3); slope = -3	
Two Points (-4, -7) and (8, -13)	

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
Parallel Lines 	**Definition:** _____ **Algebraically, how do we know if two lines are parallel?**
Perpendicular Lines 	**Definition:** _____ **Algebraically, how do we know if two lines are perpendicular?**
What are Negative Reciprocals?	**Some examples:** **1)** $\frac{3}{4}$ & **2)** 2 & **3)** $-\frac{7}{8}$ & **4)** 1 & **5)** 0 &
Given Ordered Pairs	**Determine if segments AB and CD are parallel, perpendicular, or neither:** **1.** $\overline{AB}$ formed by (-2, 3) and (2, 6); $\overline{CD}$ formed by (-1, 0) and (3, 3) **2.** $\overline{AB}$ formed by (0, 2) and (5, 4); $\overline{CD}$ formed by (1, 8) and (3, 3) **3.** $\overline{AB}$ formed by (-1, 8) and (2, 6); $\overline{CD}$ formed by (-1, 2) and (3, 3) **4.** $\overline{AB}$ formed by (2, 3) and (-1, 4); $\overline{CD}$ formed by (-5, 3) and (-4, 6)

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	5. $\overline{AB}$ formed by (0, -2) and (0, 7) $\overline{CD}$ formed by (3, -5) and (6, -5)	**6.** $\overline{AB}$ formed by (-4, 7) and (-2, 6) $\overline{CD}$ formed by (2, -2) and (-8, 3)
	7. $\overline{AB}$ formed by (3, 1) and (3, -4) $\overline{CD}$ formed by (-4, 1) and (-4, 5)	**8.** $\overline{AB}$ formed by (-3, 8) and (3, 2) $\overline{CD}$ formed by (7, 1) and (5, -1)
Given Equations	**9.** $y = 7x + 2$ and $y = 7x - 1$	**10.** $y = \frac{4}{5}x - 8$ and $y = -\frac{5}{4}x + 3$
	11. $y = -\frac{1}{3}x + 2$ and $y = \frac{1}{3}x$	**12.** $x + 6y = 30$ and $3y = 18x - 6$
	13. $5x - y = 4$ and $y = -\frac{1}{5}x + 7$	**14.** $3x - y = 2$ and $12x - 4y = 4$
	15. $y = x + 3$ and $y = -x - 5$	**16.** $y = 6$ and $x = -1$

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
WRITING Parallel & Perpendicular **EQUATIONS**	• **What are we doing?** Given a linear equation and a certain point, you can create ANOTHER equation that passes through this point and is either parallel or perpendicular to the given line. • **Keep in mind the following points:** ➢ Parallel lines have _____ slopes. ➢ Perpendicular lines have _____ _____ slopes.
PARALLEL EXAMPLES	**1.** Write the equation of the line that passes through the point **(-2, 7)** and is **PARALLEL** to the line $y = -4x + 1$.
	2. Write the equation of the line that passes through the point **(3, -1)** and is **PARALLEL** to the line $x - 3y = 9$.
PERPENDICULAR EXAMPLES	**3.** Write the equation of the line that passes through the point **(4, 3)** and is **PERPENDICULAR** to the line $y = 2x - 4$.
	4. Write the equation of the line that passes through the point **(-5, 1)** and is **PERPENDICULAR** to the line $5x + 3y = -21$.

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MORE PRACTICE

Directions: Write an equation passing through the point that is **PARALLEL** to the given equation.

5. (-4, -1); $y = 2x + 4$

6. (8, 3); $y = -\frac{1}{4}x + 7$

7. (4, 5); $x - 2y = 14$

8. (-6, 7); $5x + 2y = 10$

Directions: Write an equation passing through the point that is **PERPENDICULAR** to the given equation.

9. (3, -3); $y = \frac{3}{4}x + 5$

10. (-3, -2); $y = x - 2$

11. (2, 3); $2x + 10y = 20$

12. (-1, -6); $x + 3y = 6$

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LINEAR EQUATIONS WORD PROBLEMS

Type 1	SLOPE-INTERCEPT	Notes:

1. Evan is going to the county fair this weekend. The admission to the fair is $5 and the cost per ride is 50¢. If his parents gave him $20, write and solve a linear equation to find how many rides he can go on.

2. While visiting Crimson Lake, Sally decided to go kayaking. The rangers charge $8.50 per hour in addition to a $25.00 deposit to rent the kayak. If she rented the kayak from 11:30 a.m. to 2:30 p.m., write and solve a linear equation to find the total cost to rent the kayak.

3. Alex bought a new truck for $42,935. According to the dealer, the truck will depreciate approximately $4,200 per year. Write and solve a linear equation to find how many years until the car is worth $5,135.

4. If you buy a car wash at the gas station for $6.00, the cost per gallon is $2.35. If you have $40, write and solve a linear equation to find the number of gallons of gas you can afford.

Type 2	STANDARD FORM	Notes:

5. Sam ordered two hamburgers and three hotdogs from the concession stand at the baseball game. His bill came to $19.05. If hamburgers cost $5.25 each, write and solve a linear equation to find the cost of each hot dog.

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6. Tickets at a school play cost $4 in advance or $5 at the door. Total ticket sales for an evening production were $440. If no tickets were sold in advance, write and solve a linear equation to find the how many were sold at the door.

7. Karen's piggy bank has all dimes and quarters. The total value of the money in her piggy bank is $11.40. If Karen has 36 quarters, write and solve a linear equation to find the number of dimes she has.

8. The Oakville High School Marching Band sold gift wrap to earn money for a band trip. Solid color gift wrap sells for $3.50 per roll and printed gift wrap sells for $5 per roll. The band needs to earn $2,500 for an upcoming competition. If the band sells 325 rolls of printed paper, write and solve a linear equation to find the number of solid paper they must sell.

Type 3

POINT & SLOPE

Notes:

9. At Eagle Bay, it costs $12 per hour to rent a canoe. Nate and his friends rented a canoe for 4 hours and paid $68. Write and solve a linear equation to find the cost to rent the canoe for 7 hours.

10. A construction company charges $18 per hour for debris removal, plus a one-time fee for the use of the trash dumpster. The total fee for 8 hours of service was $219. Write and solve a linear equation to find the one-time fee for the trash dumpster.

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11. A company offers premium cable for $39.95 plus a one-time setup fee. The total cost for setup and 6 months of service is $264.70. Write and solve a linear equation to find the total cost for 2 years of service.

12. A home security company provides security systems for $5 per week, plus an installation fee. The total fee for 12 weeks of service is $210. Write and solve a linear equation to find the cost of the installation fee.

Type 4

TWO POINTS

Notes:

13. To surf the internet for 15 minutes at an airport, it costs $4.05. For 40 minutes, it costs $5.80. Write and solve a linear equation to find the cost for surfing the web for one hour.

14. Water boils at 100° Celsius or 212° Fahrenheit. Water freezes at 0° Celsius or 32° Fahrenheit. If the weather forecaster says it will be 25° Celsius today, write and solve a linear equation to find what Fahrenheit temperature this is.

15. The value of a car decreases at a constant rate. After 3 years, the value of the car is $15,000. After 2 more years, the value of the car is $11,000. Write and solve a linear equation to find the value of the car after 8 years.

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16. Customers of a satellite dish company must pay for installation of the dish, and a monthly fee for service. After 6 months, Ryan had paid $594. After 11 months, he had paid $989. Write and solve a linear equation to find the cost for installation.

NOW... YOU DETERMINE THE TYPE!

17. Kate bought $23.40 worth of two types of bird seed. Thistle bird seed sells for $1.60 per pound and wild bird seed sells for $0.95 per pound. If she bought 12 pounds of wild bird seed, write and solve a linear equation to find the amount of thistle bird she purchased. seed purchased.

18. Jeff is keeping track of his weight over several weeks. After 2 weeks, he weighs 194 pounds. After 6 weeks, he weighs 186 pounds. Write and solve a linear equation to find Jeff's weight after 12 weeks.

19. A landscape supply business charges $35 to deliver mulch. The cost of the mulch is $29 per cubic yard. Write and solve a linear equation to find the cost of having 6 cubic yards of mulch delivered to a site.

20. Hallie used a birthday check her mother gave her to start a savings account. She plans to save $25 per month as well. After 9 months, Hallie has $300. Write and solve a linear equation to find the original check amount from Hallie's mother.

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